

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – MOCK INTERVIEW (20 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization(BL5)	Assessment:	Assessment Tool 2 – Mock Interview
Weightage:	20% of CIA	Total Marks:	20 (20 Questions × 1 Mark)
CO4 Statement:	<i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i>		
Student Name:	RENEE VINCENT ASR	Reg. No.:	192424161
Section / Batch:	BTECH AI & DS	Date:	17-08-2026

Code of Conduct

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 20 questions, each carrying 1 mark. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	No. of Questions	Marks
Scatterplots & Bubble Charts	Scatterplots and bubble charts	2	2
Correlation Analysis	Correlogram and feature selection	2	2
Dimensionality Reduction Techniques	PCA, t-SNE, UMAP	3	3
Density & Overplotting Techniques	Hexbin plots and overplotting handling	2	2
Network Visualization	Nodes and edges in network graphs	2	2
Time Series Analysis	Moving average, detrending, smoothing, seasonality, noise	5	5
Geospatial Visualization	Geospatial plots, map projections, choropleth maps, spatial heatmaps	4	4
GRAND TOTAL:		20 Marks	

Annexure A – Mock Interview

1. What is the purpose of a histogram?
2. How does a box plot help identify outliers?
3. What information does a violin plot provide?
4. What is the difference between a bar chart and a histogram?
5. Why are heatmaps useful in data analysis?
6. What does color intensity represent in a heatmap?
7. What is a pair plot used for?
8. How can a density plot help understand data distribution?
9. What is meant by dimensionality reduction?
10. Why is PCA commonly used before visualization?
11. What is a principal component?
12. How does UMAP preserve data structure?
13. What is the main goal of t-SNE visualization?
14. What does a contour plot represent?
15. How are contour lines interpreted in a contour plot?
16. What is a radar chart used for?
17. What does each axis in a radar chart represent?
18. How does a parallel coordinates plot visualize multivariate data?
19. What is overplotting in data visualization?
20. Name one visualization technique used for high-dimensional data.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough and accurate understanding of concepts	Shows good understanding with minor errors	Basic understanding with limited clarity	Poor understanding or incorrect responses		
Critical Thinking & Response	50%	Provides logical, confident, and well-structured responses	Gives relevant responses with minor gaps	Responses are partially correct with limited analysis	Unable to analyse or respond effectively		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

MOCK VIVA TEST REPORT

Assessment Tool 2 – Mock Interview

Course Code	DSA06
Course Title	Data Handling and Visualization
Assessment	Assessment Tool 2 – Mock Interview
Total Questions	20
Maximum Marks	20
Marks Obtained	20 / 20
Percentage	100%
Overall Grade	Excellent

1. Assessment Overview

The mock viva was conducted using the uploaded Assessment Tool 2 – Mock Interview for DSA06 – Data Handling and Visualization. The assessment contains 20 questions, each carrying 1 mark, for a total of 20 marks. The questions covered histograms, box plots, violin plots, bar charts, heatmaps, pair plots, density plots, dimensionality reduction, PCA, UMAP, t-SNE, contour plots, radar charts, parallel coordinates, overplotting, and high-dimensional visualization.

2. Overall Performance

Metric	Result
Total Questions	20
Maximum Marks	20
Marks Obtained	20
Percentage	100%
Full-Mark Responses	20
Partially Correct Responses	0
Incorrect Responses	0
Performance Level	Level 4 – Exemplary

3. Topic-wise Performance

Topic Area	Performance
Histograms & Distribution	100%
Box Plots & Violin Plots	100%
Heatmaps & Pair Plots	100%
Dimensionality Reduction	100%
Contour Plots	100%
Radar & Parallel Coordinates	100%
Overplotting	100%
High-dimensional Visualization	100%

4. Evaluation Summary

All 20 responses were evaluated as fully correct, relevant, and clearly expressed. No incorrect or partially correct responses were recorded.

5. Strengths

- Excellent conceptual understanding of data handling and visualization.
- Accurate definitions and explanations of visualization techniques.
- Strong understanding of dimensionality reduction methods such as PCA, UMAP, and t-SNE.
- Clear understanding of distribution-based visualizations including histograms, violin plots, and density plots.
- Strong knowledge of contour plots, radar charts, and parallel coordinates.
- Correct identification and explanation of overplotting.
- Answers were concise, relevant, confident, and well structured.

6. Areas for Improvement

No major conceptual weakness was identified during this mock test. For further improvement, continue practicing brief examples or applications for each visualization technique so follow-up viva questions can be answered with both a definition and an example.

7. Rubric-based Assessment

Criterion	Assessment
Understanding of Concepts (50%)	Level 4 – Exemplary
Critical Thinking & Response (50%)	Level 4 – Exemplary
Overall Performance	Level 4 – Exemplary

8. Faculty-style Remark

The student demonstrates thorough and accurate understanding of data handling and visualization concepts. The responses are logical, confident, relevant, and well structured. The student is able to explain the purpose and interpretation of major visualization techniques and dimensionality reduction methods. Overall performance is exemplary.

9. Final Recommendation

Final Result: 20 / 20 (100%) – Excellent. The student is well prepared for the Assessment Tool 2 Mock Interview.